

SIMATS ENGINEERING
ASSESSMENT TOOL 2: Case Study

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED

CO3: *Implement cloud-based solutions using cloud storage, web APIs, and platform services for realworld applications. (BL3)*

Assessment Tool Weightage: 20%

Dave's Taxonomy: Precision (Level 3)

Total Marks: 30

Question Count: 3

Code of Conduct

I Ismail M (192524128) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action. Signature of the student: Ismail M(192524128)

Assessment Tasks Case Study 1. Case Study 3: Online Food Ordering Platform

A startup builds a cloud-based food ordering system accessible via web browsers and mobile clients. Frontend clients interact with backend services through RESTful Web APIs hosted on a cloud platform. Cloud storage is used to store menu images, invoices, and customer data securely. The infrastructure uses virtual machines and managed Kubernetes for scalability and reliability. Security is enforced through authentication tokens, HTTPS, and role-based access control. The system demonstrates integration of clients, APIs, storage, and platform services in real time.

2. 1. Compute & Infrastructure (Scalability & Reliability)

- **Virtual Machines (VMs):** Likely used for hosting legacy services, monolithic background tasks, database instances, or specialized stateful processing units that require dedicated hardware allocation.
- **Managed Kubernetes:** Used for hosting containerized microservices. A managed control plane handles routine maintenance (upgrades, node patching), while Kubernetes automates:
 - **Horizontal Pod Autoscaling (HPA):** Dynamically scaling pods based on CPU/memory utilization or custom metrics.
 - **Self-Healing:** Automatically restarting failed containers and replacing non-responsive nodes.

3. 2. Security Enforcement

- **Data in Transit (HTTPS):** Uses TLS/SSL to encrypt all network traffic between clients, APIs, and microservices, preventing eavesdropping and Man-in-the-Middle (MitM) attacks.

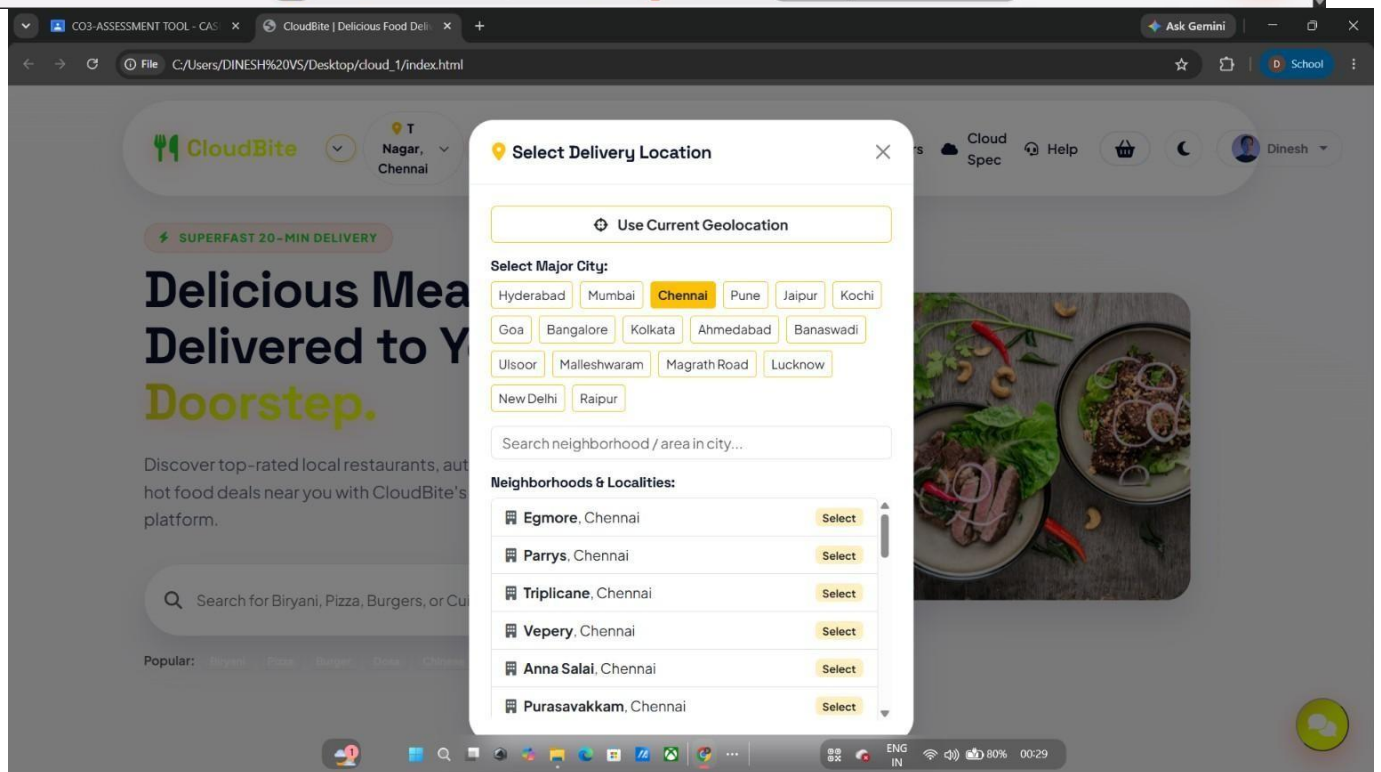
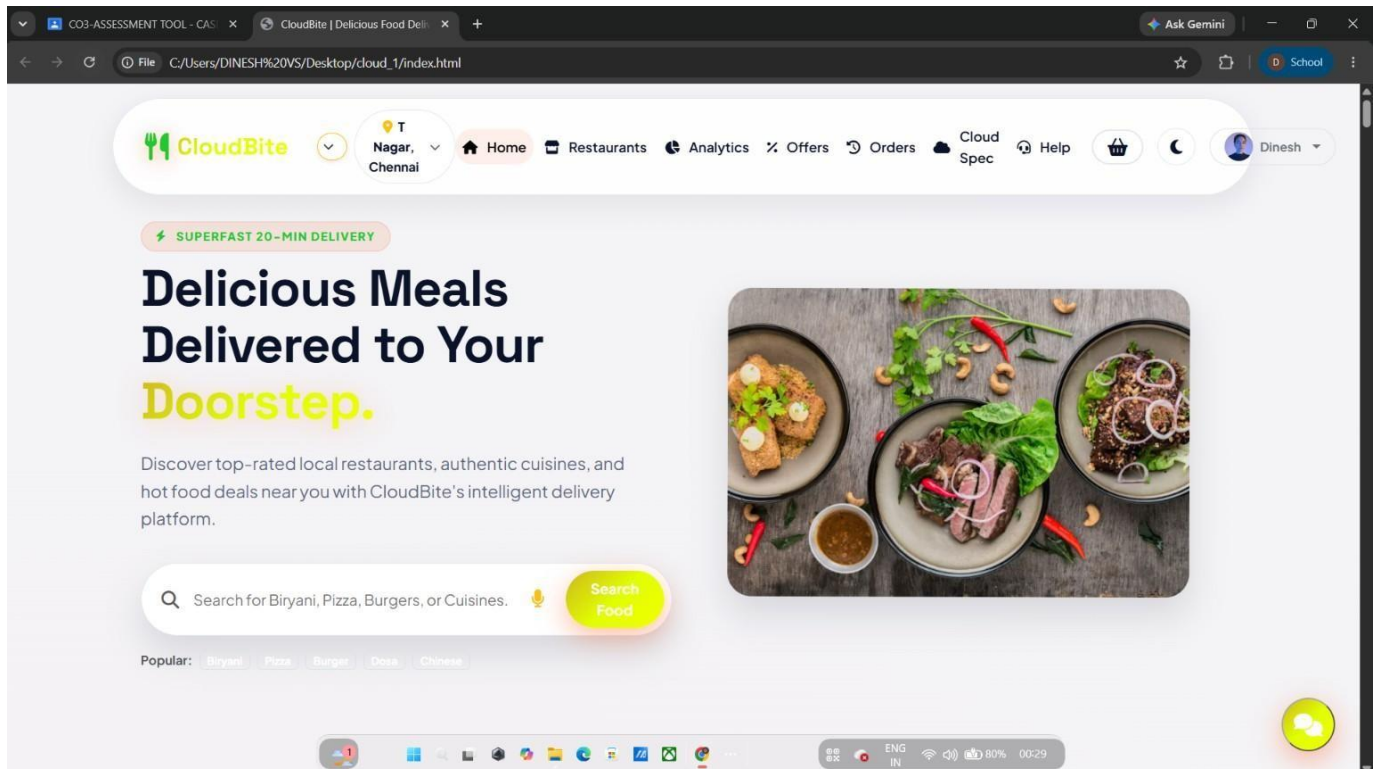
- **Authentication Tokens (e.g., JWT/OAuth2):** Stateless user authentication. Once a client logs in, tokens accompany subsequent API requests, eliminating the need to query authentication servers repeatedly.
- **Role-Based Access Control (RBAC):** Restricts system access based on user roles (e.g., *Admin*, *User*, *Auditor*) at both the application level and the Kubernetes cluster level.

4. 3. Real-Time System Integration

- **Client Layer:** Mobile apps, web frontends, or third-party client integrations.
- **API Layer (API Gateway / REST / WebSockets):** Serves as the central entry point, handling routing, rate limiting, and protocol translation.
- **Storage Services:** Combines relational databases (e.g., PostgreSQL) for structured transactions, NoSQL (e.g., MongoDB, DynamoDB) for document storage, and Object Storage (e.g., S3) for media/files.
- **Platform Services:** Asynchronous event streaming (e.g., Apache Kafka, RabbitMQ) or real-time message brokers enabling low-latency state synchronization across the system.

5. Typical Real-World Applications

6. This architecture is the industry-standard blueprint for systems requiring **high throughput, non-stop uptime, and tight security**, such as:
 - **SaaS Platforms:** Multi-tenant applications serving thousands of concurrent users.
 - **E-Commerce Applications:** Handling sudden traffic spikes during flash sales using autoscaling.
 - **FinTech & Banking Backends:** Requiring strict compliance, granular RBAC, and encrypted realtime transactions.
 - **IoT & Telemetry Platforms:** Ingesting real-time streaming data from thousands of connected devices.



CO3-ASSESSMENT TOOL - CAS

All Restaurants | CloudBite

Ask Gemini

C:/Users/DINESH%20VS/Desktop/cloud_1/restaurants.html

School



T Nagar, Chennai

HomeRestaurantsAnalyticsOffersOrdersCloud SpecHelp

 Dinesh

Explore 900+ Partner Restaurants

Powered by 123,657 real food items & restaurant metrics across 17 cities.

Pure Veg Only4.0+ Rating

Filters

SEARCH RESTAURANT / ITEM

Dish, restaurant, area...

CUISINES

☒ All Cuisines

☐ Beverages & Shakes

☐ Pizza & Fast Food

☐ Biryani & Mughlai

☐ Chinese & Sichuan

☐ North Indian

☐ South Indian

☐ Desserts & Ice Cream

☐ Street Food & Rolls

☐ Seafood



FLAT 2,750 OFF

Dindigul Veetu Kalyanam
Biryani Specialist in Egmore
Egmore, Chennai • Biryani
★ 3.8 Dining: 3.8 (0 votes)
30-53 mins ₹,1393 for two 31+ items



₹,120 OFF on orders above ₹,250

Anjappar
Beverages Specialist in Parrys
Parrys, Chennai • Beverages
★ 3.8 Dining: 3.1 (74 votes)
20-53 mins ₹,1550 for two 202+ items



FLAT 50% OFF

Mathsya
Shake Specialist in Egmore
Egmore, Chennai • Shake
★ 4.2 Dining: 4.2 (0 votes)
27-43 mins ₹,1359 for two 242+ items

Order Now

CO3-ASSESSMENT TOOL - CAS

Dataset Analytics & CSV Explorer

Ask Gemini

C:/Users/DINESH%20VS/Desktop/cloud_1/analytics.html

School



Chennai

HomeRestaurantsAnalyticsOffers



CSV DATASET ENGINE

Zomato Food & Restaurant Analytics

Interactive visual exploration of 123,657 real dataset records across 907 restaurants and 17 Indian cities.

Browse Restaurants



123,657

Total Food Dish Records



907

Partner Restaurants



17

Indian Metros & Cities



48

Unique Cuisines

 Dataset City Distribution & Partner Outlets

CO3-ASSESSMENT TOOL - CASOffers & Coupons | CloudBite

Ask Gemini

C:/Users/DINESH%20VS/Desktop/cloud_1/offers.html

School

CloudBite

HomeRestaurantsOffersOrders

EXCLUSIVE DEALS

CloudBite Food Coupons & Savings

Save big on every food order with instant discounts, bank cashbacks, & free delivery codes.

₹,100 OFF

SAVE100

Flat ₹,100 discount on orders above ₹,1399

Copy Code

50% OFF

FIRST50

Get 50% off up to ₹,1150 on your first order

Copy Code

FREE DELIVERY

FREEDELIVERY

Zero delivery fee on orders above ₹,1299

Copy Code

20% OFF

BITE20

20% Instant discount on Gourmet restaurants up to ₹,1120

Copy Code

CO3-ASSESSMENT TOOL - CASTrack Order | CloudBite

Ask Gemini

C:/Users/DINESH%20VS/Desktop/cloud_1/orders.html

School

CloudBite

HomeRestaurantsOrders

LIVE ORDER TRACKING

Order #CB-948201

Order More Food

Arriving in approximately 18 Mins at T Nagar, Chennai.

Live Delivery Progress

Order Confirmed

Received by Royal Palace Biryani at 16:42

Preparing in Kitchen

Chef is preparing your Special Mutton Biryani

Picked Up by Partner

Delivery executive assigned

Out For Delivery

Rider on the way to your destination

Delivered

Delivery Executive

Ramesh Kumar

TVS Apache (TN 01 AB 9942) • 4.9

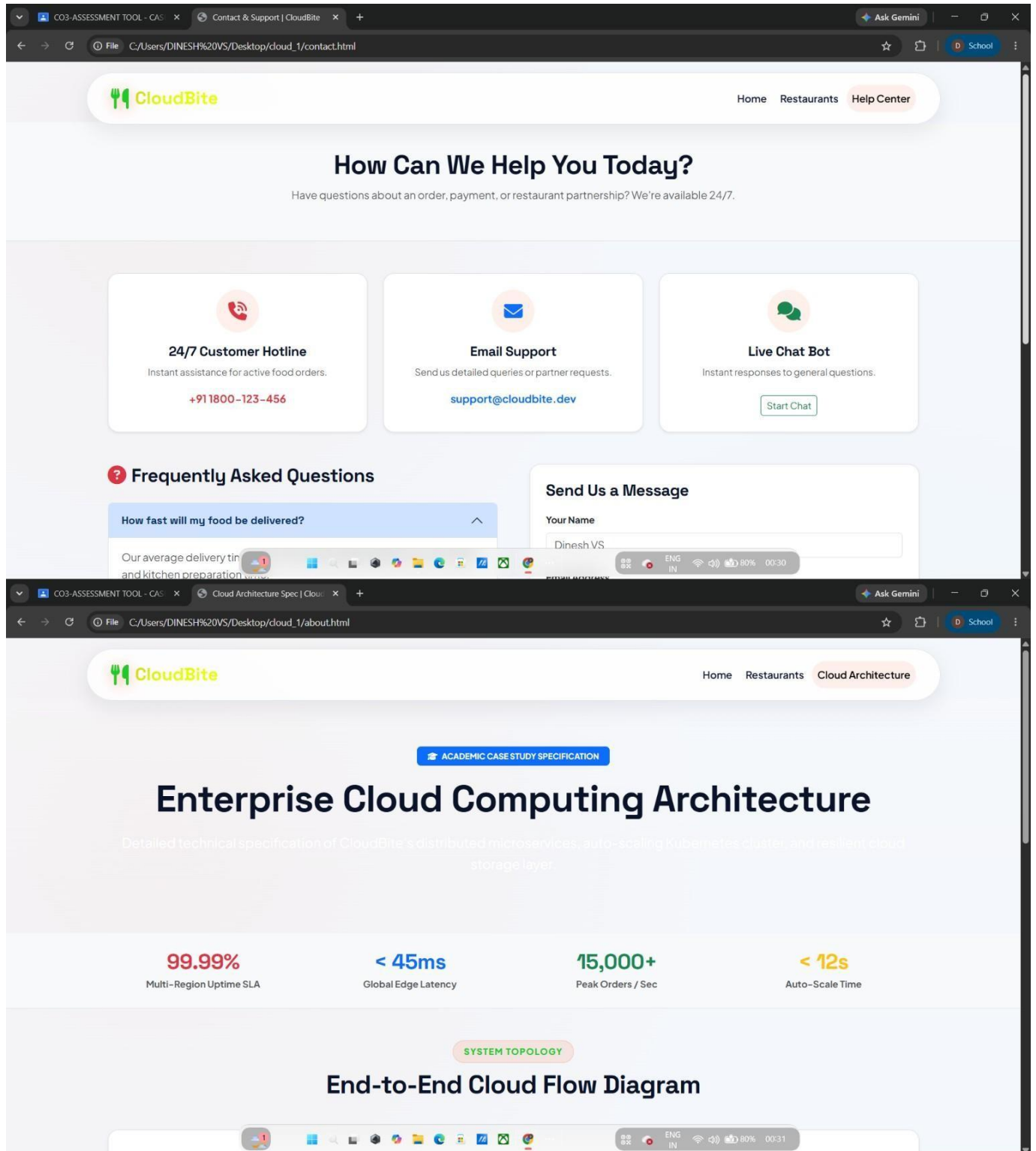
Order Details

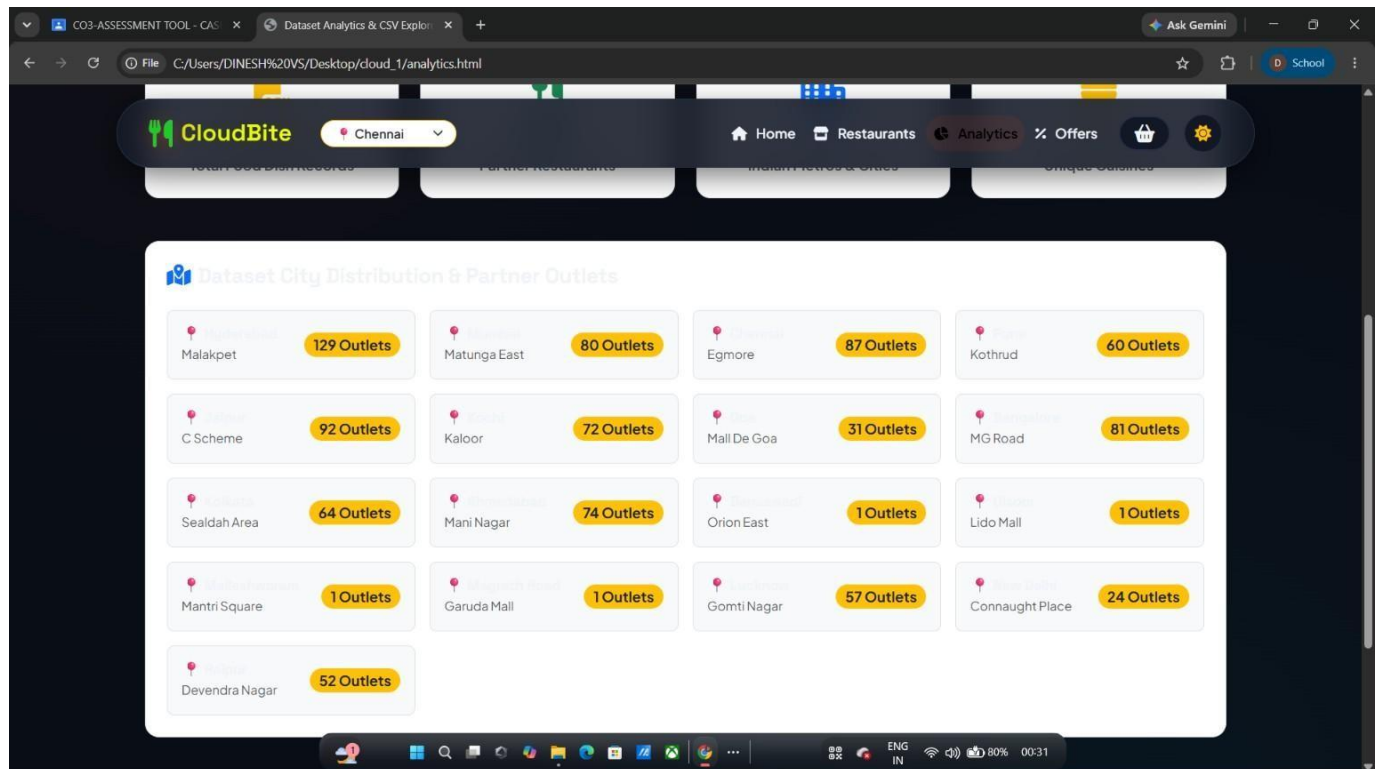
Special Hyderabad Mutton Biryani x 1

₹340

Paid via UPI

₹262





Case Study Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Understanding of Case	25%	Clearly identifies all technical and business requirements	Identifies most requirements	Basic understanding only	Poor understanding of case
Application of Concepts	25%	Applies suitable cloud concepts and tools effectively	Good application with minor issues	Partial application	Weak or incorrect application
Analysis	20%	Strong analysis with clear trade-off reasoning	Reasonable analysis	Limited analysis	Minimal analysis
Solution Design	20%	Solution is feasible, practical, and well justified	Reasonable solution with minor gaps	Basic solution	Unclear solution

Clarity	10%	Wellstructured and easy to follow	Mostly clear	Somewhat unclear	Poorly organized
---------	-----	---	--------------	---------------------	---------------------